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RESEARCH ARTICLE

Enhancing Crowd Safety at Hajj: Real-Time Detection of Abnormal Behavior Using YOLOv9

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ABSTRACT This study introduces an innovative approach for detecting abnormal behavior in real-time during the Hajj pilgrimage, utilizing the advanced capabilities of the YOLOv9 algorithm. Faced with the challenges of monitoring dense crowds, existing methodologies often fall short in accuracy, efficiency, and cost-effectiveness. Leveraging deep learning, this research accurately identifies features of abnormal behavior from the HAJJv2 dataset, specifically curated and annotated for the Hajj context. Optimization of the YOLOv9 algorithm for this scenario demonstrated superior performance metrics (mean Average Precision (mAP@0.5), Recall, and Precision) when compared with its predecessors (YOLOv4, YOLOv5, YOLOv7, and YOLOv8), highlighting significant improvements in detection accuracy and real-time applicability. A pivotal aspect of this research was the implementation of critical data pre-processing steps to enhance the object detection model's performance. These included a few-shot data sampling framework and comprehensive data augmentation techniques (flipping, rotation, scaling, and blurring) to increase the training dataset's variability and representativeness. Such pre-processing measures are instrumental in addressing occlusions and variations in viewpoints, thereby ensuring the model's robust generalization capabilities in real-world scenarios. The application of these methodologies not only improves safety and security measures during the Hajj but also offers a versatile framework that can be adapted for abnormal behavior detection in various crowded environments. This paper contributes to the field of crowd management and safety by providing a scalable, accurate, and efficient solution for the real-time detection of potential risks, ensuring a safer environment for one of the largest human assemblies worldwide.

INDEX TERMS YOLOv9, Hajj, abnormal behavior detection, deep learning, crowd management, data pre-processing, data augmentation.

I. INTRODUCTION

In the realm of computer vision, analyzing human activities in crowded scenes stands out as one of the most daunting challenges. Delving into crowd behavior remains an intricate problem in this field, drawing growing interest from vision communities. A major obstacle lies in pinpointing anomalies within densely populated areas, requiring a keen understanding of both spatial and temporal dimensions. This specific challenge revolves around the detection of irregularities in crowded environments, demanding advanced techniques to discern patterns amid the complexity of densely packed spaces and the passage of time [1]. Typically, abnormal behavior is characterized as atypical actions exhibited by an individual within a specific event, including running, backward walking, and jumping. The perception of individual abnormal behaviors can vary significantly based on the context of the crowd and the specific scenes in which they occur. Different crowd contexts and scenes may influence how these behaviors are interpreted and identified, adding complexity to the task of analyzing abnormal activities in diverse environments. The definition of abnormal behavior can vary considerably across different locations or situations. Similarly, the density and size of a crowd can vary widely, from small gatherings to massive crowds, depending on the specific circumstances. These variations underscore the

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complexity of studying crowd behavior in diverse settings, highlighting the need for adaptable and context-aware approaches in computer vision to effectively analyze and identify abnormal activities across different environments and crowd scenarios. Visual technology presents promising avenues for crowd safety. Devices such as phone cameras, CCTV systems, and satellites can serve as data collection tools. Nevertheless, computer vision encounters difficulties in crowded environments, particularly when confronted with heavy occlusions. Occlusions, which refer to situations where objects or individuals are partially or completely hidden from view, pose a significant challenge in crowded environments. In such scenarios, it becomes difficult for computer vision systems to accurately detect and track individuals, leading to potential limitations in analyzing crowd behavior and identifying abnormalities. Despite these challenges, ongoing advancements in computer vision algorithms and technologies aim to address these issues, working towards more effective solutions for crowd safety in various complex environments [2]. Researchers continue to explore innovative methods to improve the accuracy and reliability of visual data analysis, even in the presence of heavy occlusions, to enhance crowd safety measures. To enhance safety in public places, numerous studies have addressed the challenge of detecting abnormalities in crowd scenes. To enhance security in public areas, numerous research efforts have tackled the challenge of identifying anomalies in crowd environments. These studies have examined a diverse array of trajectory features Zhang et al. [3], Bera et al. [4], Zhao et al. [5], Coşar et al. [6], dense motion features Colque et al. [7], Qasim and Bhatti [8]), spatial-temporal features (Guo et al. [9], Fradi et al. [10]), as well as utilizing deep learning-based feature extraction and optimization techniques for anomaly detection (Sikdar and Chowdhury [11], Mehmood [12], Bansod and Nandedkar [13]). Many of these approaches concentrate on detecting anomalies at the frame level in a binary manner. Several studies have been dedicated to identifying irregularities in crowd surveillance footage (Bansod and Nandedkar [14], Bansod and Nandedkar [13]), with limited attention given to multi-class anomalies (Sikdar and Chowdhury [15]). This research introduces a real-time model designed for crowd behavior analysis, presenting an integrated solution capable of performing precise localization and classification of crowd anomalies into multiple distinct classes. In developing the detection model, we employed YOLOv9, a state-of-the-art single-stage object detection model known for its high accuracy and speed in real-world applications. YOLOv9's ability to detect objects in real time makes it particularly well-suited for our task of identifying abnormal behaviors within crowds, such as running and walking against the crowd. In this study, we assess the effectiveness of the proposed methods on datasets characterized by extremely high crowd density, such as the HAJJ dataset.

The Hajj, a yearly religious pilgrimage held in Makkah, Saudi Arabia, is a major large-scale gathering, consistently attracting over two million participants from various nations and regions around the world. The sheer diversity and cultural variations among pilgrims pose challenges in comprehending their abnormal behaviors. In response, this research meticulously defines, annotates, and labels a set of abnormal behaviors contextualized within the Hajj setting. The exploration of abnormal behaviors is grounded in an exhaustive study, linking them to potential obstacles or hazards affecting the flow of large-scale crowds.

Building upon the groundwork laid by Alafif et al. [2], The primary objective of this analysis is to facilitate the automated detection, tracking, and recognition of abnormal behaviors within the extensive crowds captured by surveillance cameras, ensuring the safety and seamless movement of pilgrims during Hajj. Moreover, it assists law enforcement and decision-makers in visualizing and anticipating possible threats related to crowd dynamics. This paper introduces an abnormal behavior detection system, utilizing the HAJJv2 dataset, based on YOLOv9 specifically designed for Hajj scenarios. The system architecture is detailed, and the presented experimental results demonstrate its effectiveness in detecting abnormal behaviors, ensuring the safety and smooth movement of pilgrims during the Hajj pilgrimage.

The primary contribution of this paper is:

- **Advanced Anomaly Detection for Hajj Scenarios:** Introducing an improved abnormal behavior detection system, utilizing the HAJJv2 dataset, based on YOLOv9, specifically designed for Hajj scenarios. Detailing the system architecture and presenting experimental results showcasing its effectiveness in detecting abnormal behaviors during the Hajj pilgrimage.
- **Accuracy Improvement with YOLOv9:** Enhancing the accuracy of abnormal behavior detection in the Hajj dataset by leveraging the capabilities of the YOLOv9 model. Capitalizing on the strengths of deep learning algorithms to identify nuanced abnormal behaviors that may be challenging for traditional image processing methods.
- **Real-time Detection in Hajj Context:** Leveraging the real-time detection capabilities of YOLOv9, is particularly beneficial in the dynamic and crowded environment of the Hajj pilgrimage. Addressing the need for quick and timely detection of abnormal behaviors to ensure the safety and smooth movement of pilgrims during Hajj.
- **Versatile Application to Hajj Dataset:** Demonstrating the versatility of the proposed YOLOv9-based abnormal behavior detection approach beyond Hajj scenarios. Modifying the system to identify additional objects of interest within the Hajj dataset, helping create a flexible tool for a range of applications related to crowd monitoring and safety.
- **Reduced False Alarms in Hajj Settings:** Mitigating false alarms during abnormal behavior detection in the Hajj dataset through the use of YOLOv9. Capitalizing on deep learning to discern specific features associated

with abnormal behaviors, thereby reducing unnecessary emergency responses and costs.

- **Cost-effective Implementation with Hajj Dataset:** Implementing the proposed abnormal behavior detection system using YOLOv9 with a focus on cost-effectiveness, particularly relevant for applications in the Hajj dataset. Utilizing low-cost cameras and hardware to minimize the financial burden associated with traditional detection methods during the Hajj pilgrimage.
- **Utilization of Large and Diverse Hajj Dataset:** Utilizing a large and diverse Hajj dataset containing real-world images and videos depicting various abnormal behaviors. Employing YOLOv9 on this extensive dataset to enhance precision in abnormal behavior predictions and reduce overfitting issues specific to the Hajj context.

The structure of the remaining work is outlined as follows. Section II provides an overview of recent research in abnormal behavior detection and recognition within the literature. The proposed framework is detailed in Section III. An experimental evaluation is presented in Section IV, and Section V concludes this endeavor.

II. RELATED WORK

One of the primary objectives of crowd behavior analysis is the prediction of unusual phenomena, crucial for ensuring the peaceful organization of events and minimizing casualties in public areas. This section provides a comprehensive summary of scientific literature on the topic, exploring diverse approaches grounded in various principles and methodologies.

In the realm of crowd behavior analysis, traditional methods have historically leaned on the extraction of handcrafted features, serving as the backbone for expert systems and neural networks. Notably, texture analysis has been instrumental in identifying regular and near-regular patterns in images [16]. Building on this foundation, Saqib et al. [17] employed texture descriptors for crowd density estimation, showcasing the versatility of texture-based approaches.

In contrast, a subset of methods has diverged from traditional feature extraction, delving into the realm of physics concepts and fluid dynamics for crowd analytics [18]. However, the application of these methods faces challenges inherent in the nonlinearities present in real-world images and videos, necessitating efficient handling for accurate results [19].

Recent developments in crowd behavior analysis have seen a shift towards sophisticated methodologies. Solmaz et al. [20] introduced a method based on linear estimation, employing a Jacobian matrix, optical flow, and particle advection to detect large-scale unusual crowd behaviors. Subsequent studies, such as those by Wang et al. [21] and Alqaysi and Sasi [22], employed clustering techniques and motion history images with segmented optical flow, respectively, to extract features and detect abnormal behaviors.

Advancements continued with Zou et al. [23] identifying extensive crowd movements and pathways using trackless association, a technique later adopted by Bera et al. [4] who examined irregular behavior patterns through Bayesian learning methods. Pennisi et al. [24] advanced the method by segmenting the features obtained to detect unusual crowd behaviors. Recently, Fradi et al. [10] and Wu et al. [25] concentrated on examining extensive crowd characteristics through visual feature descriptors, reflecting the progress in the field of crowd behavior analysis. Alafif et al. [2] introduce two hybrid approaches that combine Convolutional Neural Networks (CNNs) and Random Forests (RFs) for detecting and recognizing unusual behaviors in crowd videos of varying scales. These methods are developed and evaluated using a meticulously annotated and labeled Hajj dataset (HAJJv2), designed to cover a diverse range of large-scale crowd abnormal behaviors.

Certain computer vision-based methods have emerged to address the challenges posed by nonlinearities, particularly by scrutinizing groups of people exhibiting coherent movements [26]. Other approaches center on path analysis using mathematical models, while insights from psychologists contribute to understanding emergency and situational awareness [27]. A common theme unifying these methods is the heightened demand for security measures and the monitoring of crowded environments, uncovering applications such as person tracking, anomaly detection, behavior pattern analysis, and context-aware crowd counting.

Despite the integration of deep learning solutions boasting high accuracy rates, persistent challenges mar the field of crowd analysis. These challenges include density variation, irregular object distribution, occlusions, and pose estimation [28]. Addressing these challenges remains paramount for the continued enhancement of crowd analysis methodologies.

Deep learning (DL)-based methods commonly demand extensive data for training, verification, and testing, leading to challenges such as false regressions and increased computational overhead due to the large scale of the datasets. In an effort to overcome these issues, a comprehensive dataset is being curated, with intentions for public accessibility in the near future. Convolutional Neural Networks (CNNs): CNNs excel in image processing tasks, making them suitable for abnormal behavior detection in Hajj scenarios by recognizing common features such as color and texture. Their advantages include simplicity, high accuracy with large datasets, and real-time detection. However, they may face challenges in complex backgrounds or low-light conditions and require substantial training data.

Recurrent Neural Networks (RNNs): Designed for sequential data, RNNs enhance abnormal behavior detection in videos by capturing temporal dependencies. They are versatile but computationally intensive, may have difficulty handling long-term dependencies, and require substantial amounts of training data. YOLOv2: Utilizing a single neural network, YOLOv2 predicts bounding boxes and class probabilities simultaneously, achieving high accuracy and real-time processing. Its simplicity is an advantage, but it may face challenges in complex backgrounds or low-light conditions and require substantial training data.

YOLOv3: An upgraded version of YOLOv2, YOLOv3 features a more advanced architecture, capable of managing challenging backgrounds and dim lighting conditions. Despite requiring more computational resources, it achieves high accuracy in real-time abnormal behavior detection.

YOLOv4: Building on the YOLO algorithm, YOLOv4 introduces enhanced features for better abnormal behavior detection, effectively managing challenging backgrounds and dim lighting conditions. However, it requires greater computational resources than its predecessors.

YOLOv5: The latest version, YOLOv5, employs a lightweight architecture and introduces features like autoanchor optimization. Faster and more accurate than YOLOv4, it offers state-of-the-art performance on multiple benchmarks. Yet, it requires substantial training data and may face challenges in detecting small or complex-shaped objects.

YOLOv8: YOLOv8, developed by Ultralytics—the same team behind the influential YOLOv5 model—features several architectural updates and improvements compared to YOLOv5.

It's important to acknowledge that although each model offers distinct advantages, there are trade-offs that need to be thoughtfully considered based on particular application requirements. Some current research gaps and constraints of these models are:

CNNs may encounter difficulties detecting abnormal behaviors in images with complicated backgrounds or inadequate lighting. RNNs tend to be computationally demanding, may have trouble capturing long-term relationships, and often require extensive training data. YOLOv2 may struggle with identifying abnormalities in images with intricate backgrounds or low-light conditions and also necessitates large training datasets. YOLOv3 typically demands more computational power than YOLOv2. YOLOv4 generally requires even higher computational capacity than previous versions. YOLOv5, while effective, still needs substantial training data and may face challenges with spurious regressions and detecting small or complex-shaped objects.

Despite advancements in deep learning-based approaches for abnormal behavior detection, several challenges persist. More diverse and extensive datasets are required for effective training and evaluation of these methods. Additionally, detection algorithm performance can be hindered by low-quality cameras or poor lighting conditions.

This work presents a method based on YOLOv9 for identifying unusual behaviors in Hajj scenarios. As far as we know, this is the initial application of the YOLOv9 algorithm in this context. The proposed approach seeks to tackle challenges associated with detecting anomalies in large crowds, offering enhanced accuracy, real-time detection, adaptability, and cost-effectiveness.

In computer vision, YOLO (You Only Look Once) algorithms are widely recognized for their impressive accuracy and compact model size. YOLO models are accessible to a broad range of developers as they can be trained on a single GPU and deployed affordably on edge hardware or in the cloud. The latest and most advanced YOLO approach, YOLOv9, can be applied to various tasks, including abnormal behavior detection, image classification, and segmentation.

YOLOv9, introduced by Wang et al. [29] addresses this issue by incorporating Programmable Gradient Information (PGI), a novel technique designed to retain crucial information throughout the detection process. PGI features a reversible branch that operates in conjunction with the main network, preserving key features and enhancing training outcomes without adding computational overhead. Additionally, they introduced a Generalized Efficient Long-Range Attention Network (GELAN) block, which optimizes the balance between parameter count, complexity, accuracy, and inference speed, allowing users to select the most suitable computational blocks for various devices.

III. METHODOLOGY

A. HAJJv2 DATASET

The HAJJv2 dataset, introduced by Alafif et al. [2], includes 18 videos that were manually collected during the annual Hajj pilgrimage, stored in mp4 format. These videos, captured using high-resolution cameras, showcase individuals displaying abnormal behaviors within large crowds and depict scenes from various locations of the Hajj, such as 'Massaa,' 'Jamarat,' 'Arafat,' and 'Tawaf.' The videos are subsequently cropped and divided into training and testing sets, with each set comprising nine shorter clips. The training videos run for 25 seconds, while the testing clips have a length of 20 seconds. Abnormal behaviors shown in the dataset include actions such as standing, sitting, sleeping, running, moving against or in varying directions from the crowd, and the presence of non-pedestrian elements like cars and wheelchairs. To underscore the risks posed by these behaviors to crowd movements, Figure 1 provides examples of the abnormal actions within the dataset. For the purpose of training and testing, manual annotation and labeling of individuals' anomalous behaviors in the videos are conducted using Roboflow for faster labeling [30].

TABLE 1. Description of the dataset Hajjv2.

No.	Classes	Training	Testing
1	Different Direction	7,152	6,262
2	Opposite Direction	36,577	18,802
3	Non Pedestrian	4,186	4,146
4	Running	51	190
5	Sitting	63,383	6,446
6	Sleeping	2,400	2,618
7	Standing	19,773	14,107
Total		**133,522**	**52,571**

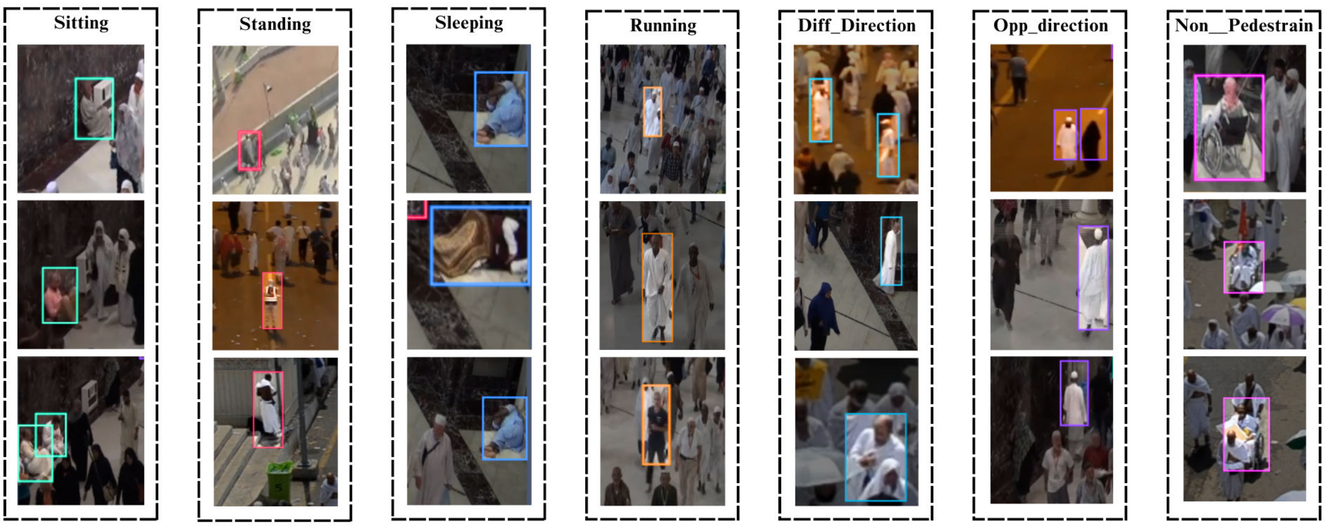

FIGURE 1. Instances of abnormal behaviors within the HAJJv2 dataset.

As represented in table 1, the training file contains 133,522 annotated abnormal behaviors, while the testing file includes 52,571 annotated abnormal behaviors.

This study took a significant step toward constructing a robust abnormal behaviors detection model in the Hajj dataset by implementing essential data pre-processing procedures. The main goal of this step was to improve the object detection model's performance. This involved automatically selecting frames that accurately represent the training data and enhancing the training dataset with sufficient variability through data augmentation. These actions were intended to help the model generalize effectively to real-world test cases. Specifically, the study carried out two key data pre-processing steps:

- **Implementing a few-shot data sampling framework:** Selectively extracting frames from individual videos assembled a training dataset that is highly representative of the diverse behaviors observed during the Hajj event. Utilizing Roboflow, we selectively extracted frames from individual videos to ensure the efficient collection of a highly representative training dataset, capturing the diverse behaviors observed during the Hajj event. Roboflow employs a dedicated process for frame extraction, capturing four frames per second of the video. This systematic approach enhances the diversity and richness of the training dataset, thereby contributing to the effectiveness of the subsequent model.
- **Data Augmentation:** In computer vision, data augmentation is a widely employed technique to enhance the sample size and diversity of training datasets. This becomes particularly crucial in object detection tasks, such as the one undertaken in our current work, where challenges like occlusion can complicate the learning process. To address issues related to occlusion and viewpoint variation, we applied various data augmentation techniques in this study.
  - 1) Flipping: Horizontal image flipping was implemented to assist the model in learning to detect abnormal behaviors from the Hajj dataset.
  - 2) Rotation: Rotational augmentation was applied to diversify the data by altering the viewpoint angle of abnormal behaviors.
  - 3) Scaling: Scaling was employed to modify the size of objects in the image, facilitating the model in learning to detect abnormal behaviors of different sizes.
  - 4) Blurring: Image blurring was performed to aid the model in learning to detect abnormal behaviors under poor lighting conditions.

In the HAJJv2 dataset, model performance is often hindered by frequent misdetections caused by severe occlusions and tracking failures resulting from changes in lighting. These variations in illumination impact the model's ability to reliably detect abnormal behaviors, thereby reducing its overall accuracy. Additionally, the high computational demands of these models lead to prolonged processing times, which limits their effectiveness in real-time applications, especially in large-scale crowd scenarios like the Hajj pilgrimage. By addressing these issues, the model's robustness and detection accuracy could be significantly improved, making it more suitable for densely populated environments.

B. OBJECT DETECTION APPROACHES

There are two main categories for object detection techniques: one-stage and two-stage [31], [32], [33]. Prominent examples of one-stage methods include YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector). These methods allow for the immediate prediction of object boundaries and category labels in a single traversal of the neural network. This approach accelerates data processing (inference time) by eliminating the step of identifying potential object regions. An example of a two-stage method is Faster

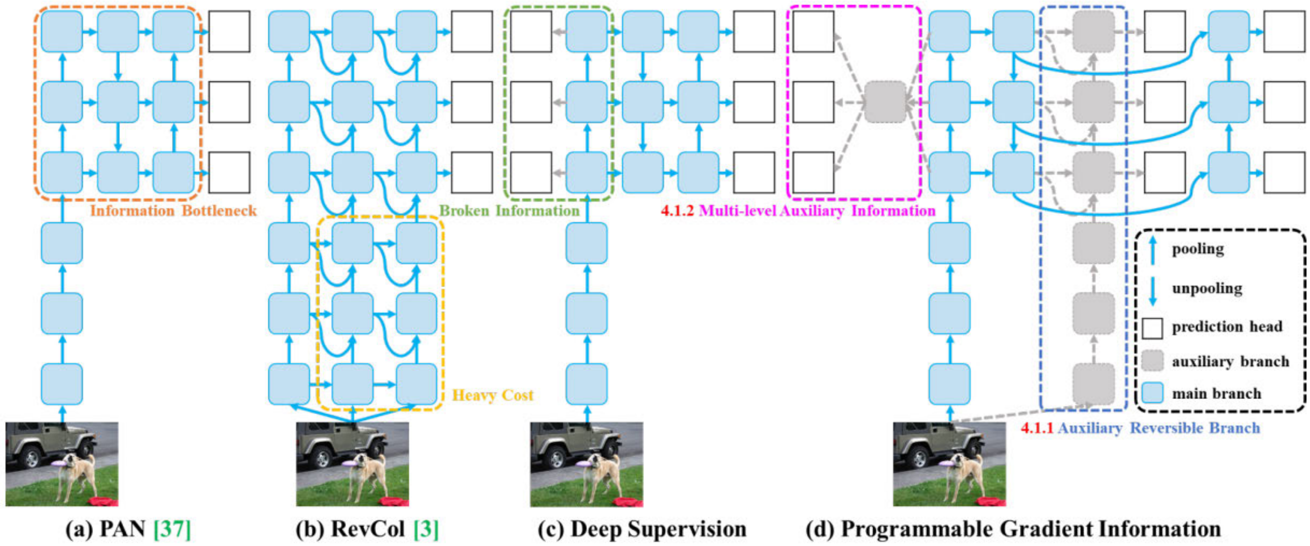

FIGURE 2. YOLOv9 architecture [29].

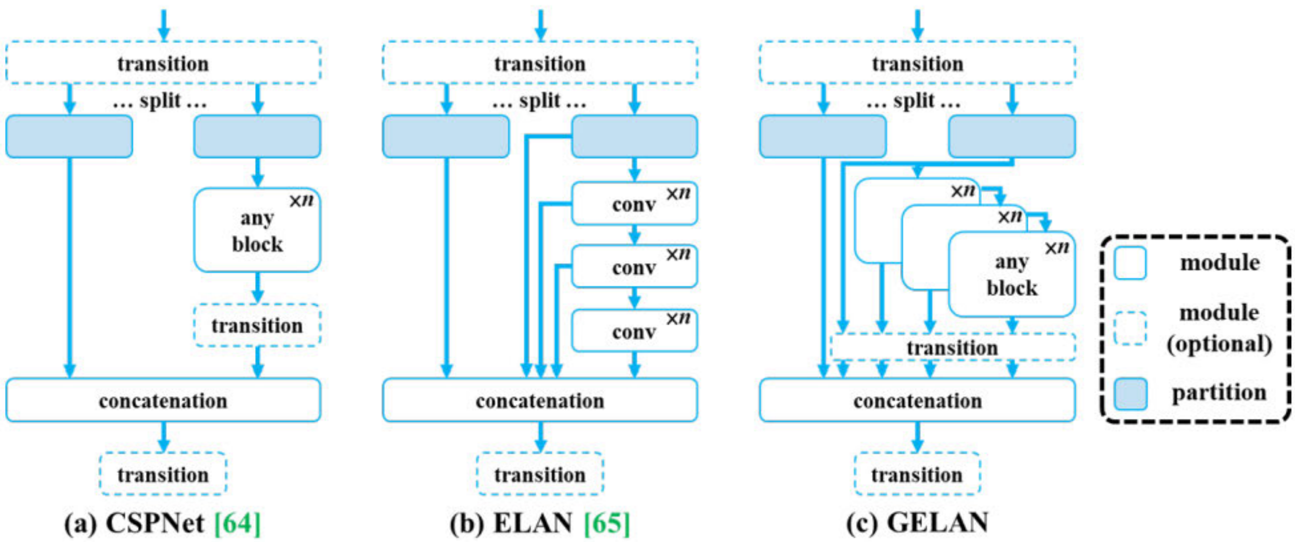

FIGURE 3. The GELAN architecture integrated into YOLOv9 [29].

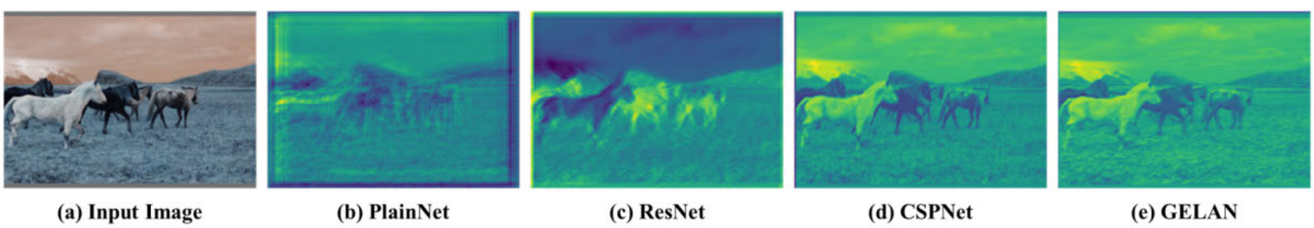

FIGURE 4. Details of output feature maps from randomly initialized weights across different deep learning network architectures [29].

R-CNN (Region-based Convolutional Neural Network) [33]. Initially, Faster R-CNN uses a specialized network called the Region Proposal Network (RPN) to generate candidate regions. These regions are further optimized in the next stage to produce precise object boundaries and category predictions. Single-stage methods prioritize efficiency and straightforward design, making them well-suited for real-time applications where quick processing is essential, such as in self-driving cars and surveillance systems. Examples of these methods include YOLO and SSD. On the other hand, two-stage methods, while generally achieving higher accuracy, involve greater complexity and longer processing

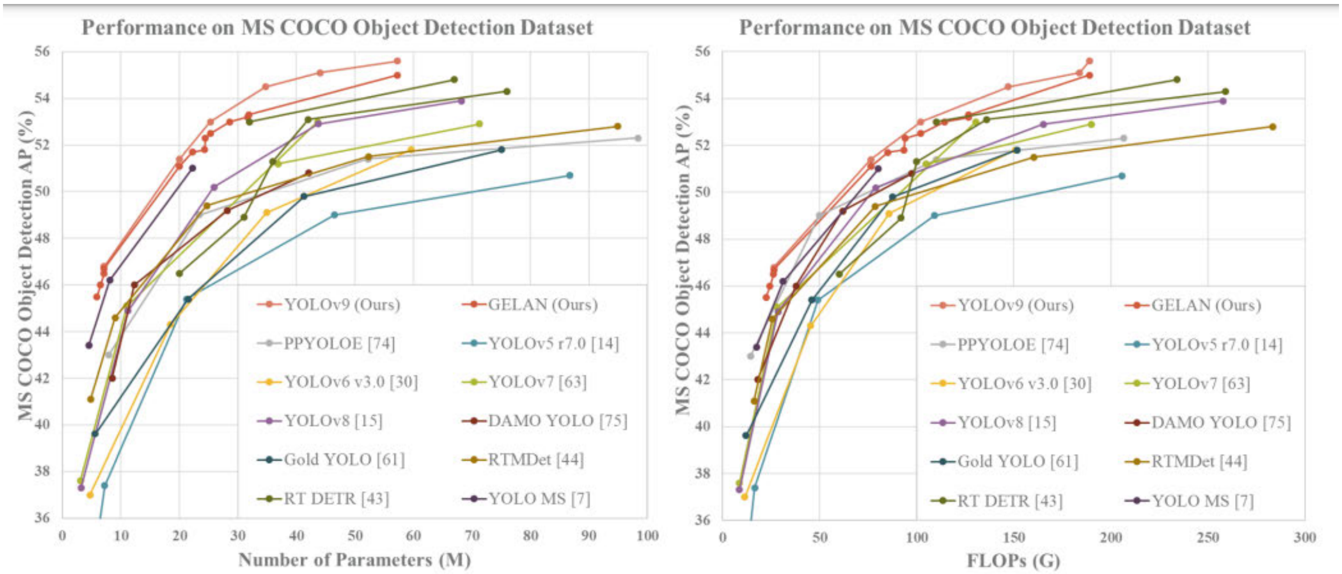

FIGURE 5. Comparison of state-of-the-art real-time object detectors with YOLOv9 [29].

times. When high accuracy is essential and there are adequate computational resources, two-stage methods like Faster R-CNN may be more appropriate.

1) YOLOv9: A CUTTING-EDGE OBJECT DETECTION MODEL

In this research, we focus on the novel design of the YOLOv9 architecture, which introduces key advancements in both efficiency and accuracy through the integration of Programmable-Gradient-Information (PGI) and Generalized-Efficient Layer Aggregation Network (GELAN). These advancements are central to the model's capability to improve real-time object detection, especially in complex and crowded environments.

Released in February 2024 [29], YOLOv9 signifies a significant leap in real-time object detection technology. The PGI design enhances the flow of gradient information throughout the network, preventing the degradation of essential data across layers. This ensures that gradient information is preserved during the backpropagation phase of training, significantly improving the learning process and enabling more accurate object recognition. The introduction of PGI directly addresses challenges associated with traditional deep learning architectures, where crucial input information may be lost or distorted.

Moreover, the GELAN architecture plays a critical role in optimizing the network's efficiency. It incorporates principles from CSPDarknet and ELAN, aiming to create a lightweight and efficient design without compromising accuracy. GELAN aggregates information across multiple layers while reducing computational load, which is crucial for maintaining high inference speeds in real-time detection tasks. The integration of GELAN within YOLOv9 is a key innovation, ensuring that the model is both accurate and efficient for complex object detection in dynamic environments.

These algorithmic innovations make YOLOv9 a significant leap forward in the field of object detection, providing improved accuracy, faster inference, and a more robust design that can handle dense, dynamic scenes such as abnormal behavior detection in crowds. The algorithm's novel structure is pivotal to the model's high performance and serves as a core contribution to the research.

2) PROGRAMMABLE-GRADIENT-INFORMATION (PGI)

In neural networks, information can diminish or be lost as it moves through layers, a challenge referred to as the information bottleneck. Gradients play a crucial role in updating network weights during training, and their accuracy is vital for successful learning. Programmable Gradient Information (PGI) addresses this issue by ensuring that gradient data is maintained consistently across the network [29]. It prevents the loss of crucial input data during backpropagation, thus improving the model's learning effectiveness and its ability to recognize objects.

3) YOLOv9's PROGRAMMABLE GRADIENT INFORMATION ARCHITECTURE

The YOLOv9 PGI architecture consists of three main components: a main branch, an auxiliary reversible branch, and multi-level auxiliary information (a detailed explanation of these components can be added here with a reference, or refer the reader to Figure 2).

4) GENERALIZED-EFFICIENT-LAYER-AGGREGATION-NETWORK (GELAN)

YOLOv9 also incorporates a novel architecture called Generalized Efficient Layer Aggregation Network (GELAN) [29]. GELAN is built upon principles derived from two well-known techniques: CSPDarknet (Cross-Stage-Partial-Network) and Efficient Layer Aggregation Network (ELAN).

Designed for lightweight operation, GELAN leverages gradient path planning to efficiently aggregate information across different layers in the network. This focus on lightweight design translates to faster inference speeds while maintaining accuracy. By directly tackling the information bottleneck issue, GELAN enhances YOLOv9's accuracy and efficiency in live object detection scenarios. The specific integration of GELAN within YOLOv9 is illustrated in Figure.3.

5) GELAN PRESERVES INPUT INFORMATION

Figure 4 visualizes the output feature maps generated by various network architectures after processing randomly initialized weights. As illustrated in figure 4, the GELAN architecture demonstrates a significant capacity to retain essential input data throughout the forward propagation.

6) YOLOv9: PUSHING EFFICIENCY AND ACCURACY BOUNDARIES

An evaluation of YOLOv9 reveals significant improvements over current state-of-the-art (SOTA) models across various performance metrics. YOLOv9 showcases better parameter efficiency, requiring fewer parameters while maintaining or surpassing the accuracy of leading models. Figure 5 illustrates a comparison between YOLOv9 and advanced real-time object detection models. This comparison underscores YOLOv9's novel integration of PGI and GELAN, optimizing the balance between efficiency and accuracy in object detection. These features make YOLOv9 the preferred model for abnormal behavior classification.

C. TRAINING AND EVALUATION

We utilized Python programming within the Google Colab environment to train and evaluate results, including validation and testing. At the time of this study, the most recent iteration of YOLO was YOLOv9. This deep-learning neural network model demonstrates improved speed and accuracy in object detection compared to its predecessors. The study employed the YOLOv9 algorithm and its pre-trained weights as a foundation for training on custom data, utilizing transfer learning. Like its predecessors, other YOLO versions, the detection of objects with a bounding box relies on six essential attributes. These attributes consist of the x and y coordinates for the top-left corner of the bounding box, the bounding box's width and height, a confidence score indicating the probability between zero and one, and a class ID. From a performance standpoint, YOLOv9 was already an excellent algorithm for object detection. This study further contributed to enhancing detection accuracy in the context of Hajj crowd scenarios. YOLOv9 stands as the latest state-of-the-art object detection model in the YOLO series, developed by Wang et al. [29] it is available for public training and testing.

The architecture of YOLOv9 is depicted in Figure 2. YOLOv9 can be broadly divided into two main functional components: a backbone network for feature extraction and a detection head for object prediction. **Backbone Network**: This initial stage extracts informative features from the input image. YOLOv9 likely leverages a convolutional neural network (CNN) architecture for this purpose. The specific details of this network (e.g., number of layers, types of convolutional blocks) would depend on the variant of YOLOv9 being used. **Detection Head**: This component takes the feature maps produced by the backbone network and processes them to predict bounding boxes and class probabilities for the objects present in the image. The YOLOv9 detection head likely employs techniques like anchor boxes and non-max suppression (NMS) for object localization and classification, similar to previous YOLO versions. For a deeper understanding, you can explore specific YOLOv9 variants and delve into the details of their backbone network architecture and detection head design. This information might be available in the research paper introducing YOLOv9 [29]. Compared to its predecessors, YOLOv9 achieves increased efficiency through a combination of factors, likely including a focus on lightweight design principles and the use of efficient building blocks within the convolutional neural network architecture.

IV. RESULTS AND DISCUSSION

A. PERFORMANCE METRICS

In this study, all experiments were performed using Python in the Google Colab environment. The YOLOv9 algorithm was employed for training through transfer learning, leveraging its base weights along with a custom dataset, specifically the HAJJv2 datasets. After training, the weights saved from this process were used to assess performance on both validation and test sets.

To thoroughly and objectively assess the model's performance, we employed the mean average precision (mAP) metric to evaluate accuracy and examine object detection results. True positives (TP) represent the count of target frames accurately identified as positive, false positives (FP) indicate the number of target frames incorrectly predicted as positive, and false negatives (FN) denote the target frames that are actually positive but misclassified as negative.

Precision, defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{1}$$

is defined as the proportion of correctly predicted positive target boxes out of all the target boxes that the model identified as positive.

Recall, defined as

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{2}$$

represents the proportion of true positives relative to the total of true positives and false negatives, offering a thorough assessment of the model's capability to identify positive instances.

Represents the ratio of true positives to the sum of true positives and false negatives, providing a comprehensive

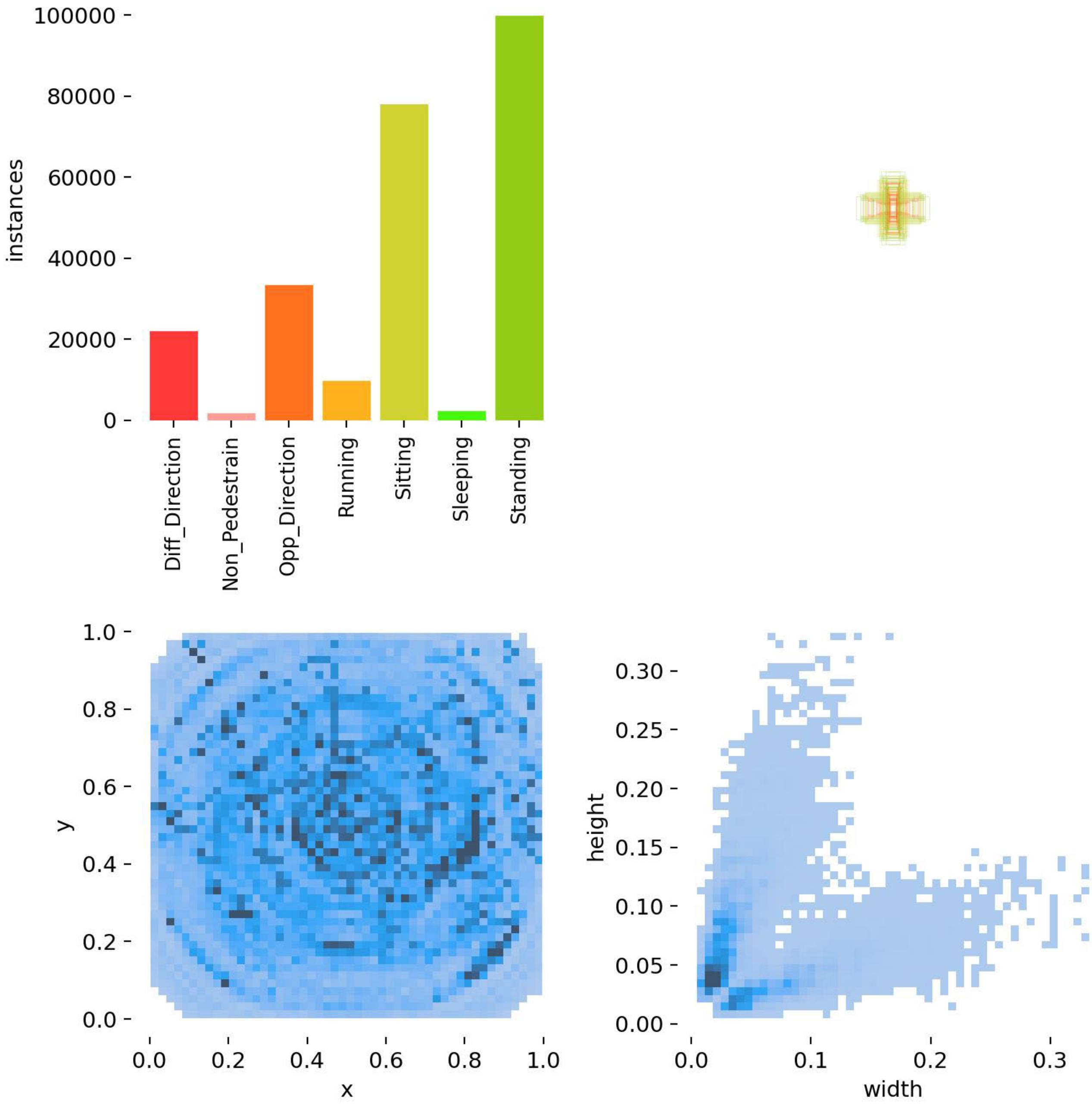

FIGURE 6. An overview of the class distribution and visualizations associated with object detection.

evaluation of the model's ability to detect positive cases. Mean average precision (mAP) is a comprehensive metric used to assess the effectiveness of object detection models across different categories. It computes the average precision (AP) for each category and then calculates the overall performance by averaging these AP values.

$$\mathrm{mAP} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{AP}_i \tag{3}$$

Here, C represents the total number of classes in the dataset. A greater mAP value signifies improved model performance.

The evaluation of object detection results was performed using mAP (mean average precision) with two distinct outcomes (mAP50 and mAP50-95), as predefined by the YOLOv9 algorithm. Bounding box overlap, measured as the intersection over union (IoU) between the ground truth and predicted bounding box, was a key metric in this

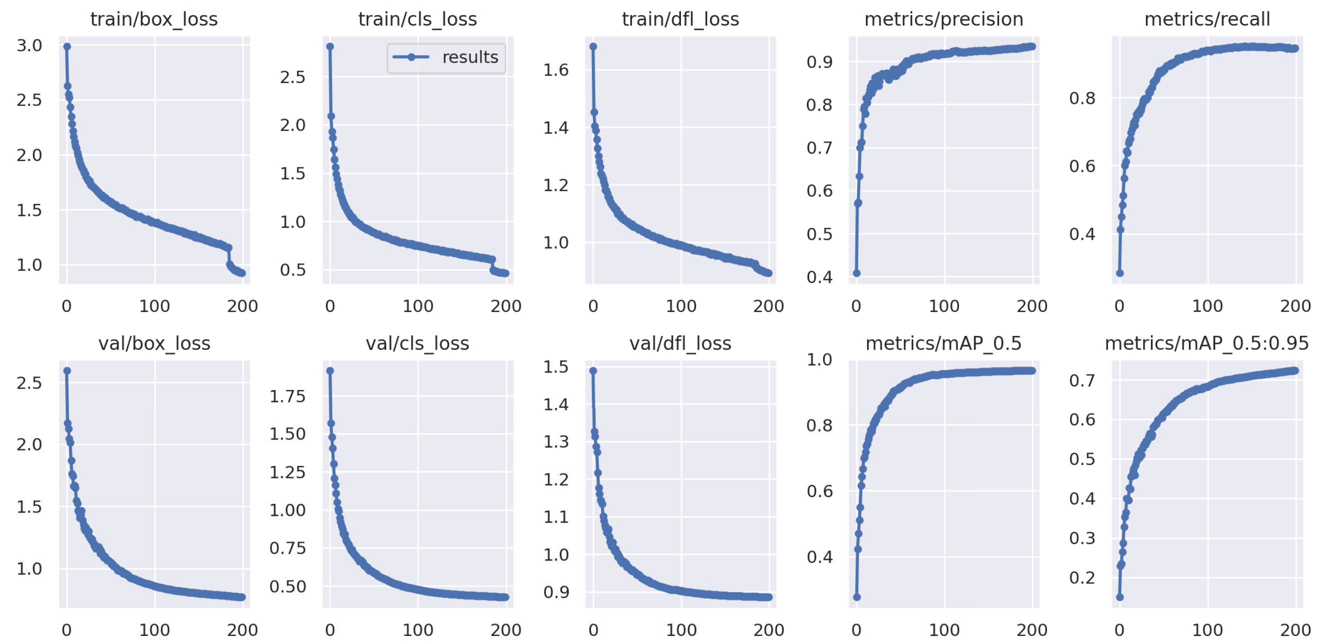

FIGURE 7. Graphs illustrating the mean Average Precision (mAP) and loss trends after training the YOLOv9 model for detecting abnormal behaviors in Hajj.

study. In mAP50, a detection was deemed true positive if the IoU exceeded 0.5. Additionally, mAP50-95 considered thresholds ranging from 0.5 to 0.95 in increments of 0.05. The primary metric employed for result presentation was mAP50, with occasional use of mAP50-95 as specified. The YOLO algorithm lacked inherent image augmentation techniques, delegating this aspect to third-party tools like Roboflow.

Roboflow played a dual role in this process. Initially, it was utilized for annotating the HAJJv2 dataset, providing valuable annotations to enhance the dataset. Additionally, Roboflow was employed to generate diverse augmented versions of the dataset, contributing to the richness and variety of the training data. This diversity is crucial for enhancing model performance and generalization. Subsequently, the training performance was assessed on the dataset to train and evaluate the overall model performance.

B. EXPERIMENTAL SETTING

The experiments in this study were conducted using a Tesla V100-SXM2 GPU with NVIDIA drivers on an Ubuntu 20.04 system. A Python 3 environment was utilized for all the experiments described in this paper.

C. TRAINING CONFIGURATION

The model was trained for a total of 300 epochs. To accommodate the limitations of the available GPU memory, a batch size of 16 was chosen. The training process leveraged the YOLOv9 model from the YOLO family with an Adam optimizer and a learning rate of 0.01. For the remaining training parameters, the default settings from the YOLOv9 implementation were used.

TABLE 2. YOLOv9 object detection results using HAJJv2 dataset.

Class	Instances	mAP50	Precision	Recall
All	22845	0.964	0.936	0.942
Diff_Direction	1992	0.931	0.878	0.906
Non_Pedestrian	157	0.964	0.920	0.948
Opp_Direction	3179	0.963	0.943	0.919
Running	890	0.933	0.883	0.911
Sitting	7347	0.981	0.977	0.956
Sleeping	177	0.995	0.997	1.000
Standing	9103	0.979	0.952	0.952

D. QUANTITATIVE RESULTS

In this section, we perform an in-depth evaluation of our proposed model, drawing comparisons with state-of-the-art methods. The experiments revolve around the HAJJv2 dataset, employing YOLOv9 as the benchmark.

The results from the YOLOv9 object detection model using the HAJJv2 dataset demonstrate robust performance across various classes of abnormal behaviors as shown in table 2.

With an overall mAP50 of 0.964, a precision of 0.936 indicating a reduction in false positives, and a recall of 0.942 demonstrating an improved ability to identify true positives, the model effectively detects and classifies abnormal activities in crowded scenes.

Specifically, the model shows high accuracy in detecting "Sitting" and "Sleeping" behaviors with mAP50 values of 0.981 and 0.995, respectively, and perfect recall for "Sleeping" (1.000). Classes such as "Diff_Direction" and

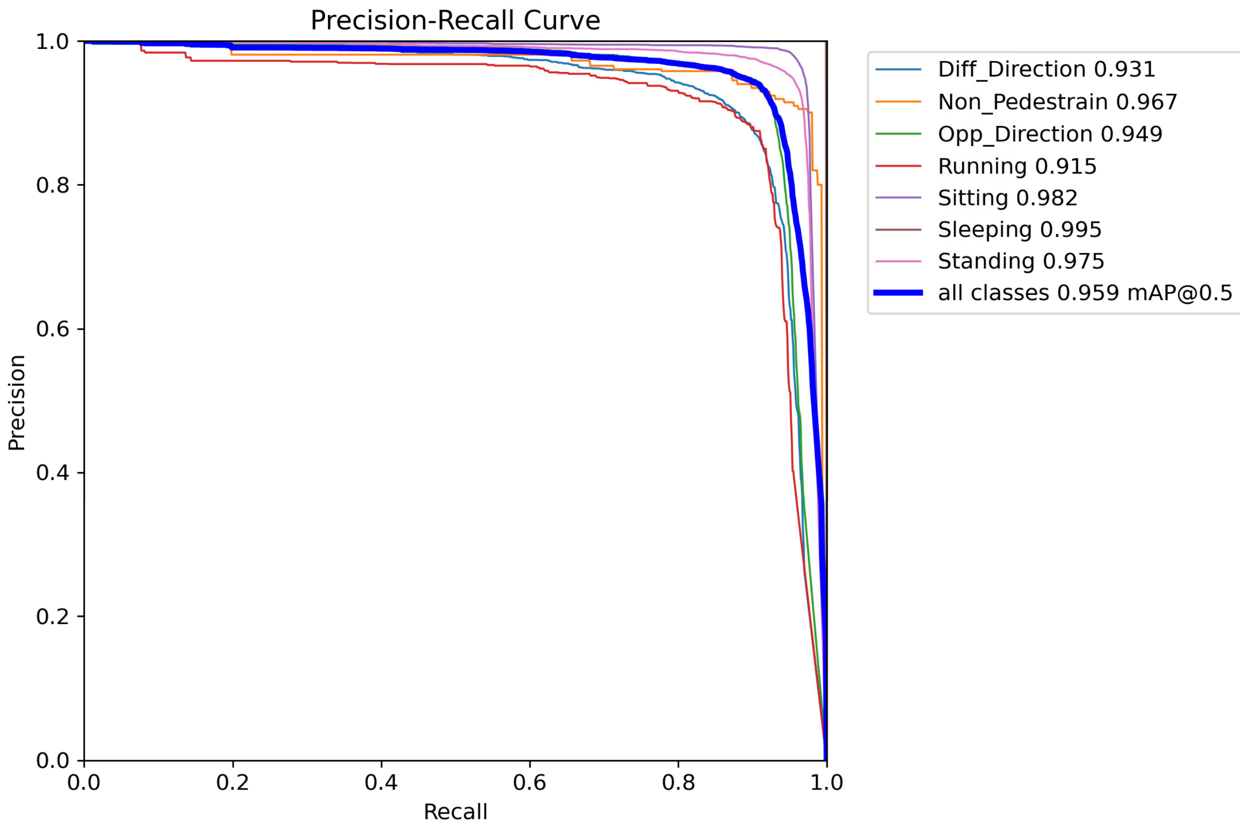

FIGURE 8. Precision-recall curve.

"Running" exhibit slightly lower precision (0.878 and 0.883, respectively), indicating the presence of some false positives, but maintain strong recall values (0.906 and 0.911), ensuring the majority of true instances are identified. The "Non_Pedestrian" and "Opp_Direction" classes also perform well, with precision and recall values above 0.90. These results underscore the model's capability to provide accurate real-time detection of various abnormal behaviors, crucial for maintaining crowd safety during the Hajj pilgrimage.

TABLE 3. Performance metrics of different YOLO models using the HAJJv2.

Model	mAP@0.5	Recall	Precision
YOLOv4	0.786	0.875	0.536
YOLOv5	0.807	0.756	0.784
YOLOv7	0.899	0.863	0.853
YOLOv8	0.912	0.868	0.899
YOLOv9	**0.964**	**0.948**	**0.939**

Table 3 presents a comprehensive overview of object detection metrics, encompassing various YOLO models—YOLOv4, YOLOv5, YOLOv7, YOLOv8, and YOLOv9—evaluated on mAP@0.5, recall, and precision values using the HAJJv2 dataset. In terms of mAP@0.5, YOLOv9 emerges as the leading model, boasting the highest value of 0.964, signifying superior accuracy in object detection compared to its predecessors. When evaluating recall performance, YOLOv9 demonstrates a commendable recall rate of 0.948, showcasing its ability to capture a substantial portion of true positive instances. YOLOv4 and YOLOv7 also exhibit robust recall values, while YOLOv5 slightly trails behind. Turning to precision metrics, YOLOv9 achieves the highest precision score of 0.939, striking a harmonious balance between true positive detections and minimizing false positives. YOLOv7 and YOLOv8 also display high precision, whereas YOLOv4 and YOLOv5 exhibit lower precision scores. In summary, YOLOv9 stands out as the exemplary model, excelling in mAP@0.5, recall, and precision. Its well-balanced precision-recall trade-off positions it as a robust choice for accurate and reliable object detection across crowded scenarios. The experimental outcomes underscore the strong performance of the model proposed in this paper. To visually demonstrate the enhanced accuracy of our proposed model, we have selected representative images from the test dataset.

E. QUALITATIVE RESULTS

A depiction of class distribution and related visualizations for object detection can be found in Figure 6.

Figure 7 presents ten graphs illustrating various metrics throughout the training and validation phases of the model developed for detecting abnormal behavior in Hajj using the HAJJv2 dataset. The metrics analyzed are Box Loss,

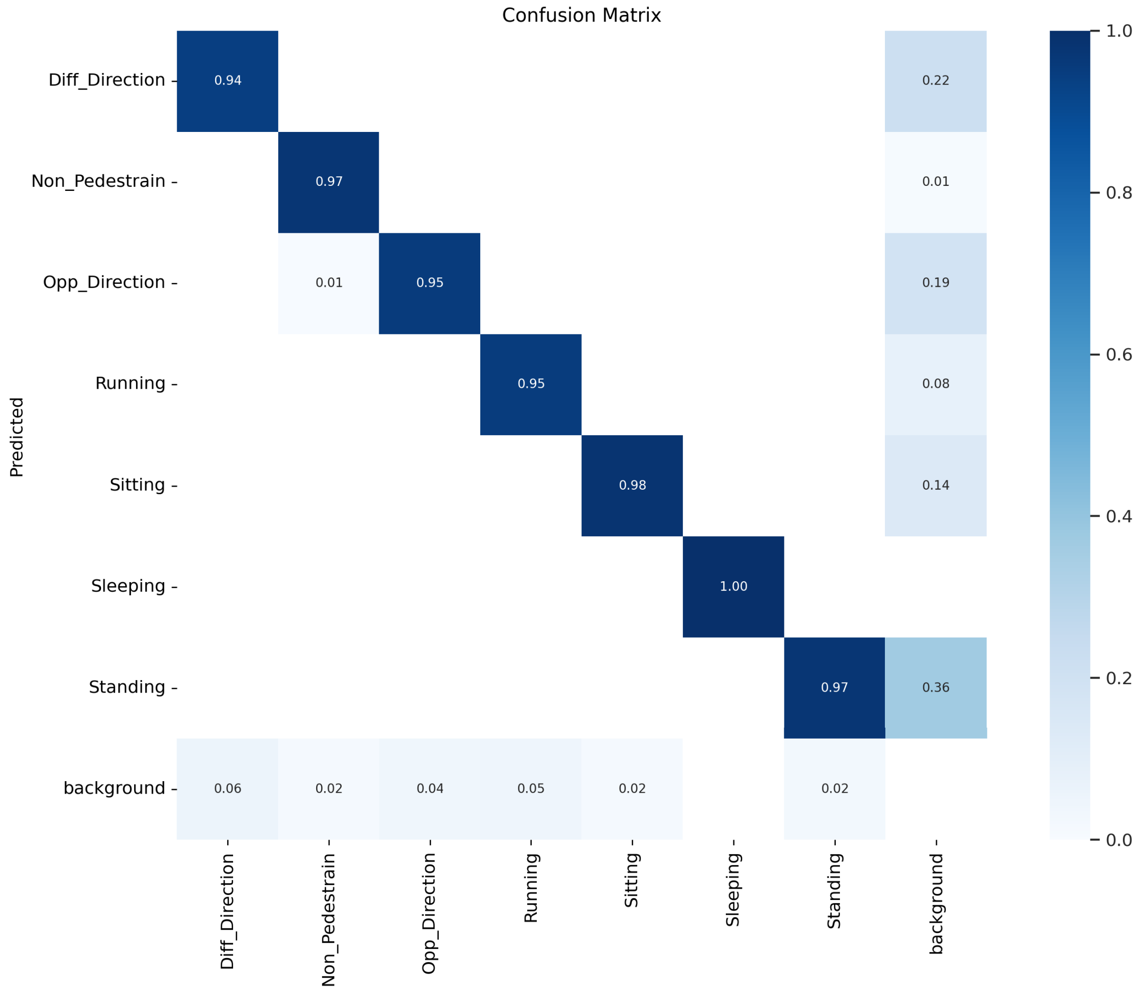

FIGURE 9. **Confusion matrix of model for detecting abnormal behavior in Hajj.**

Classification Loss, Distribution Focal Loss, Precision, Recall, and Mean Average Precision (mAP). The graphs show an overall positive trend over the epochs, with reductions in loss metrics (Box Loss, Classification Loss, Distribution Focal Loss) indicating improvements in the model's performance. Concurrently, increasing values for precision, recall, and mAP suggest enhanced performance.

The confusion matrix in Fig. 9 shows the model performs well, with high precision for most classes, particularly 'Sleeping' (1.00) and 'Non_Pedestrian' (0.97). However, there is notable confusion between 'Standing' and 'background' (0.36), indicating a need for better differentiation. While overall performance is strong, reducing misclassifications with the background could further enhance accuracy. Additionally, Figure 8 displays the precision-recall (PR) curves for the proposed model. Notably, precision shows a rapid increase as recall improves. The PR curves' proximity to the top-right corner highlights the model's high recall and precision. The large area under the PR curves suggests robust model performance. Moreover, the smoothness of the PR curves demonstrates a stable relationship between recall and precision in our model.

Figure 10 presents multiple frames from the HAJJv2 dataset, demonstrating how our model identifies and classifies different actions within a dense crowd. Each image is marked with dotted rectangles that depict various crowd behaviors, such as sitting, moving in opposite directions, standing, running, sleeping, and shifts in crowd movement direction. These visualizations showcase the model's capability to accurately identify and distinguish between different crowd behaviors in complex, densely populated environments, enhancing crowd monitoring and safety protocols during large-scale events like the Hajj pilgrimage.

The model's capacity to precisely recognize and differentiate between various crowd behaviors in such complex and highly crowded settings is further illustrated by these visualizations, which contribute to improving crowd management and safety measures during major events such as the Hajj pilgrimage.

FIGURE 10. Visuals illustrating the identification of irregular actions utilizing Yolov9.

V. CONCLUSION

This paper introduces an improved method for real-time abnormal behavior detection during Hajj, employing the YOLOv9 algorithm. Through the application of deep learning, our approach aims to identify abnormal behavior features, potentially enhancing accuracy in crowded settings. Additionally, it provides a cost-effective alternative to traditional detection methods. The proposed system employs a deep neural network trained on an extensive dataset of abnormal behavior images, achieving a high precision rate of 96.4% across all classes. Tailored for Hajj scenarios, the system leverages the HAJJv2 dataset and YOLOv9, offering improved accuracy, real-time detection, and versatility beyond Hajj contexts. By using YOLOv9, the system reduces false alarms, ensures cost-effective implementation, and maximizes precision with a diverse dataset. Addressing Hajj-specific challenges, the proposed solution provides a holistic approach to crowd monitoring and safety, extending its applications beyond abnormal behavior detection.

Future work will focus on further improving the model's generalization capabilities by incorporating more diverse datasets and analyzing its performance in different real-world environments. Additionally, a more thorough theoretical analysis will be conducted to assess the model's computational complexity, convergence behavior, and gap to optimality.

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